# KIWi | Project Details

AI-Powered Knowledge Assistant for the THWS CampusPortal

**Status:** Planning / early development | **Team:** 4 students
**Module:** MAI Project Module, Winter Semester 2026/27
**Institution:** THWS | **Supervisor:** Prof. Dr. Ivan P. Yamshchikov

## Overview and Problem

KIWi (KI-Wissensassistenz) is a proposed, locally hosted assistant for CampusPortal (HISinOne). It will use Retrieval-Augmented Generation (RAG) to turn questions into answers grounded in approved university documentation, with source citations.

CampusPortal knowledge is distributed across manuals, internal wikis, training materials and process documents. KIWi aims to make this information easier for students and staff to find and understand.

## Planned Features

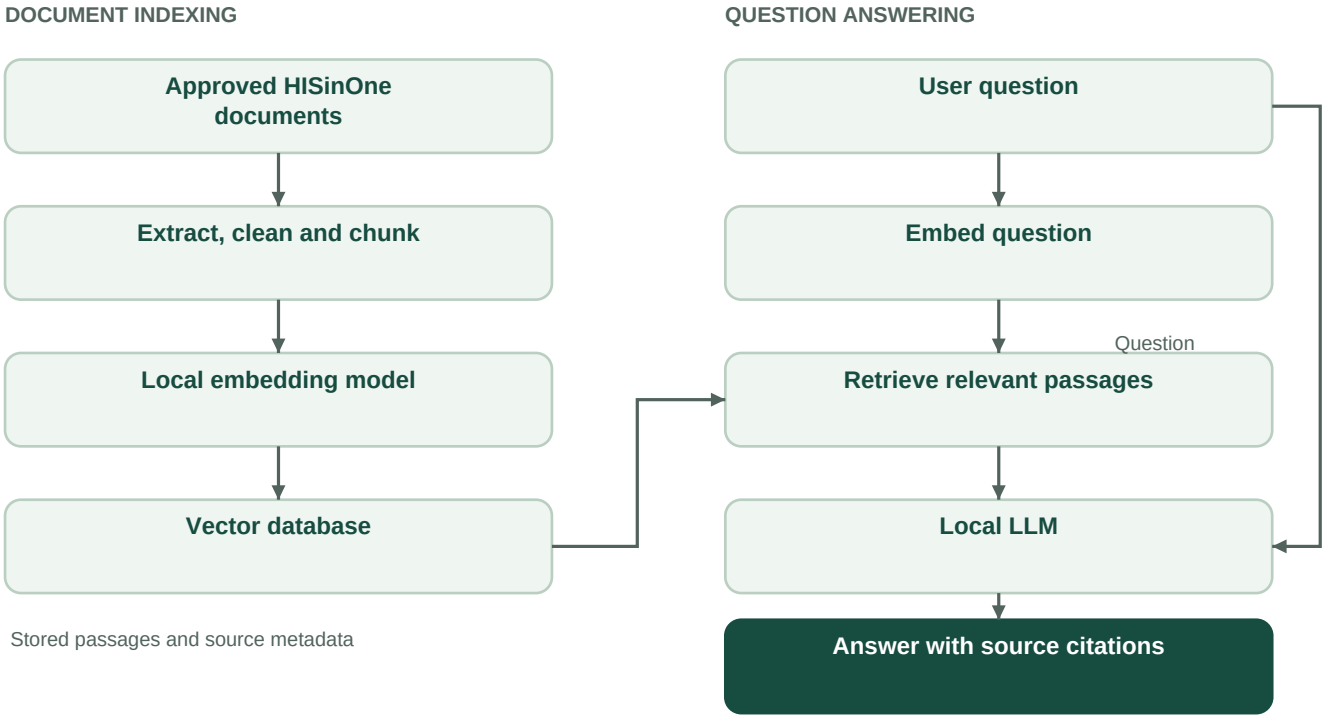
• **Natural-language Q&A:** CampusPortal guidance through a conversational interface.

• **Document ingestion and semantic retrieval:** Index approved sources and find relevant passages by meaning.

• **Grounded answers:** Local model inference with references to supporting documents.

• **Reproducible deployment:** A Dockerized prototype with quality evaluation.

## RAG Workflow




Documents are indexed locally. Each question retrieves evidence from the index; the local LLM combines that evidence with the question to produce a cited answer.

# Implementation and Delivery

Proposed technologies, milestones and project requirements

## Proposed Technology Stack

| Component | Candidate technologies |
|---|---|
| Language | Python |
| Local inference | Ollama or vLLM |
| Language model | Mistral, Llama or Qwen |
| RAG framework | LangChain or LlamaIndex |
| Vector database | ChromaDB or pgvector |
| Interface | To be selected; e.g. Streamlit or a web frontend |
| Deployment | Docker |

The final stack will be selected after requirements analysis with the THWS HISinOne team.

## Project Milestones

| Milestone | Expected content |
|---|---|
| Poster session | Requirements, knowledge audit, architecture, technology choices and pipeline plan. |
| Technical paper | RAG design, knowledge-base implementation and evaluation methodology and results. |
| Live demo | Dockerized prototype answering representative anonymized CampusPortal questions, supported by evaluation results. |

## Evaluation

Use representative anonymized queries to assess retrieval relevance, answer accuracy, citation correctness and hallucination rate. Verify that the prototype can be deployed and tested reproducibly.

## Privacy and Data Handling

Use only approved documentation and respect THWS access permissions. Document processing and model inference are intended to remain on university-controlled infrastructure without external LLM API calls. Internal documents and credentials must not be published; local hosting alone does not establish GDPR compliance.

## Deliverables

Dockerized working prototype and source repository; short technical paper; project poster; live demonstration; evaluation results.

The project remains in planning. Installation instructions and confirmed milestone dates will be added when available.